

- 16.** A musical organ produces notes by blowing air into a set of pipes that are open at one end and closed at the other.

The speed of sound in the air in the pipes is 320 m s^{-1} .

What is the lowest frequency of sound produced by a pipe of length 2.0 m ?

- A** 20 Hz **B** 40 Hz **C** 80 Hz **D** 160 Hz

Ans: B

There is a node at the closed end and an antinode at the open end.

Therefore, at the fundamental (lowest) frequency,

Length of the pipe, $L = \frac{\lambda}{4}$

Therefore, lowest frequency of sound, $f = \frac{v}{\lambda} = \frac{v}{4L} = \frac{320}{4(2.0)} = 40 \text{ Hz}$

17 The figure below shows two speakers, S_1 and S_2 , placed 2.00 m apart. The two